

Cube-Dependent Function Norms Separate Generalized BMO from BMO

Full negative answers to Questions 6.1 and 6.2 of arXiv:2204.11099

Automated proof attempt
agent_lane_03 model: GPT5.6

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Status: full counterexamples, likely valid, pending human verification. Question 6.1 fails even for Banach local spaces. Question 6.2 and its explicitly proposed stronger version both fail for quasi-Banach local spaces: we construct a normalized family $X = \{X_Q\}$ such that $BMO \hookrightarrow BMO_X$, while $x \mapsto x_1$ belongs to BMO_X^* and not to classical BMO.

Source questions and answer

For every axis-parallel cube $Q \subset \mathbb{R}^n$, the source assigns a local quasi-Banach function space X_Q with $\|\chi_Q\|_{X_Q} \simeq 1$, and defines

$$\|f\|_{BMO_X} = \sup_Q \|f - \langle f \rangle_Q\|_{X_Q}, \quad \|f\|_{BMO_X^*} = \sup_Q \inf_{c \in \mathbb{R}} \|f - c\|_{X_Q}.$$

Question 6.1 asks whether $BMO_X \hookrightarrow BMO$ can fail. Question 6.2 asks whether $BMO \hookrightarrow BMO_X$ forces that converse; its stronger version replaces BMO_X by BMO_X^* in the conclusion [1]. The appended source crop reproduces the questions.

Theorem 1 (Answers to Questions 6.1 and 6.2). *For every $n \geq 1$:*

1. *There is a normalized family of Banach function spaces X_Q and a locally integrable $f \in BMO_X \setminus BMO$.*
2. *There is a normalized family of quasi-Banach function spaces X_Q such that*

$$BMO \hookrightarrow BMO_X, \quad BMO_X^* \not\hookrightarrow BMO.$$

Indeed $f(x) = x_1$ belongs to BMO_X^ in the second construction. Consequently both the ordinary and stronger forms of Question 6.2 have negative answers.*

Proof intuition

The family in the definition may depend on the cube, with no compatibility between different cubes. For Question 6.1, a weighted L^1 norm can look only near the central hyperplane of each cube. The affine function x_1 then has bounded weighted oscillation although its ordinary oscillation grows linearly with the side length.

The additional embedding in Question 6.2 makes this harder: concentrating a fixed L^1 weight would detect the logarithmic drift of BMO averages between a large cube and its tiny centre. The remedy is a power mean with an exponent $p_Q \downarrow 0$. Such a mean approaches a geometric mean. We balance a diffuse scale L and a tiny central scale s so their weighted geometric mean is exactly constant. The average drift of any BMO function is only logarithmic, and the same vanishing exponent absorbs it uniformly.

A Banach-space counterexample for Question 6.1

Let Q have centre c_Q and side length L . Define the probability measure

$$d\nu_Q(x) = Z_Q^{-1} e^{-|x_1 - (c_Q)_1|} dx \quad (x \in Q), \quad X_Q = L^1(Q, \nu_Q).$$

These are Banach function spaces over Lebesgue measure because the densities are strictly positive; moreover $\|\chi_Q\|_{X_Q} = 1$.

Take $f(x) = x_1$. Its Lebesgue average on Q is $(c_Q)_1$. Writing $A = L/2$, integration in the first coordinate gives

$$\|f - \langle f \rangle_Q\|_{X_Q} = \frac{\int_0^A t e^{-t} dt}{\int_0^A e^{-t} dt} = 1 - \frac{A}{e^A - 1} \leq 1.$$

Thus $f \in \text{BMO}_X$. On the other hand,

$$\frac{1}{|Q|} \int_Q |f - \langle f \rangle_Q| dx = \frac{L}{4},$$

which is unbounded as $L \rightarrow \infty$. Hence $f \notin \text{BMO}$, proving the first assertion of Theorem 1.

A stronger quasi-Banach counterexample for Question 6.2

Fix $n \geq 1$, and put $L_0 = \exp(\exp(n+4))$. If a cube Q has side length $L \leq L_0$, let

$$X_Q = L^1(Q, dx/|Q|).$$

Suppose now that $L > L_0$, and set

$$a = \log L, \quad b = \log a, \quad q = \frac{1}{b}, \quad p = \frac{1}{a^2 b}, \quad s = L^{-(b-1)}. \quad (1)$$

Let J_Q be the cube concentric with Q and of side length s . Since $b > 1$, it is a proper subcube. Define

$$\mu_Q = (1-q) \frac{\chi_Q dx}{|Q|} + q \frac{\chi_{J_Q} dx}{|J_Q|}, \quad X_Q = L^p(Q, \mu_Q). \quad (2)$$

The measure is a probability and its density is positive almost everywhere on Q . Thus X_Q is a quasi-Banach function space over Q , with the lattice and Fatou properties and weak order unit χ_Q . Also $\|\chi_Q\|_{X_Q} = 1$ exactly. Hence the whole family is normalized.

We use the standard nested-average estimate. If $U \subset V$ are concentric cubes with side-length ratio at most two, then

$$|\langle h \rangle_U - \langle h \rangle_V| \leq \frac{|V|}{|U|} \frac{1}{|V|} \int_V |h - \langle h \rangle_V| \leq 2^n \|h\|_{\text{BMO}}.$$

Iterating through concentric doublings from J_Q to Q yields

$$\frac{1}{|J_Q|} \int_{J_Q} |h - \langle h \rangle_Q| \leq C_n (1 + \log(L/s)) \|h\|_{\text{BMO}}. \quad (3)$$

For definiteness, the doubling chain above permits $C_n = 1 + 2^n(1 + 1/\log 2) < 2^{n+2}$.

Lemma 2 (The forward embedding). *The family (2) satisfies $\text{BMO} \hookrightarrow \text{BMO}_X$.*

Proof. Let $N = \|h\|_{\text{BMO}}$. The assertion is immediate on cubes with $L \leq L_0$. On a large cube, the monotonicity of probability power means for $0 < p < 1$, followed by (3), gives

$$\left(\frac{1}{|Q|} \int_Q |h - \langle h \rangle_Q|^p \right)^{1/p} \leq N, \quad \left(\frac{1}{|J_Q|} \int_{J_Q} |h - \langle h \rangle_Q|^p \right)^{1/p} \leq RN,$$

where $R = C_n(1 + D)$ and

$$D = \log(L/s) = ab, \quad pD = \frac{1}{a}. \quad (4)$$

Therefore

$$\|h - \langle h \rangle_Q\|_{X_Q} \leq N((1-q) + qR^p)^{1/p}. \quad (5)$$

It remains only to see that the last factor is uniform. Since $D = ab > 1$, $1 + D \leq 2ab$. The displayed choice of C_n , together with $b \geq n + 4$, gives

$$\log R \leq \log(2C_n) + \log a + \log b \leq 3b.$$

Consequently

$$q \log R \leq 3, \quad v := p \log R \leq \frac{3}{a^2} < 1.$$

For $0 \leq v \leq 1$, $e^v - 1 \leq 2v$. Hence

$$\frac{1}{p} \log((1 - q) + qR^p) \leq \frac{1}{p} \log(1 + 2qv) \leq 2q \log R \leq 6.$$

The factor in (5) is at most e^6 , independently of Q . Taking the supremum proves the embedding. $\square$

Lemma 3 (The affine witness). *For the same family, $f(x) = x_1$ belongs to BMO_X^* , but $f \notin \text{BMO}$.*

Proof. Let $c = (c_Q)_1 = \langle f \rangle_Q$. On cubes with $L \leq L_0$, $\|f - c\|_{X_Q} = L/4 \leq L_0/4$. On a large cube, direct integration of the first coordinate gives, with $A = L/2$ and $B = s/2$,

$$\|f - c\|_{X_Q} = \frac{1}{(p+1)^{1/p}} ((1 - q)A^p + qB^p)^{1/p}. \quad (6)$$

The parameter choice makes the weighted geometric mean constant:

$$A^{1-q}B^q = \frac{1}{2}L^{1-q}s^q = \frac{1}{2}. \quad (7)$$

To compare the positive power mean with this limit, let $u = p \log(A/B) = pD = 1/a$. Factoring the right side of (6) by the geometric mean produces

$$H = e^{qu} ((1 - q) + qe^{-u}).$$

The elementary inequalities $e^{-u} \leq 1 - u + u^2/2$ and $\log(1 + t) \leq t$ imply

$$\log H \leq qu + \log(1 - qu + qu^2/2) \leq \frac{qu^2}{2}.$$

Using (1),

$$\frac{qu^2}{2p} = \frac{1}{2}.$$

Since $(p+1)^{-1/p} < 1$, equations (6)–(7) give

$$\|f - c\|_{X_Q} \leq \frac{e^{1/2}}{2}$$

on every large cube. The supremum over all cubes is finite, and choosing this c in the best-constant infimum proves $f \in \text{BMO}_X^*$. Finally, the classical mean oscillation is $L/4$ on every cube of side L , so $f \notin \text{BMO}$. $\square$

Lemmas 2 and 3 prove the second assertion of Theorem 1. Notice that the same witness also lies in BMO_X , because the chosen constant is its Lebesgue cube average.

Scope, checks, and literature boundary

The first construction strengthens Question 6.1 by using Banach spaces. The second fully answers Question 6.2 as written and refutes its strongest stated variant. It uses exponents $p_Q < 1$; the analogous Banach-only version of Question 6.2 is not claimed here. Fixed weighted L^1 norms, intersections, and normalized sums with $\exp L$ were also examined, but either fail the forward embedding or lose uniform normalization.

The script `code/verify_parameters.py` evaluates the exact power means stably at scales as large as $\log L = 10^6$. It verifies $D = ab$, $pD = 1/a$, the geometric balance (7), and generous forms of both uniform estimates. The proof above is analytic and has no computational dependency.

The source appeared in *Mathematische Annalen* in 2024. Exact-ID, exact-question, citation, and core-keyword searches through 11 August 2026 found no prior matching counterexample. The directly related later preprint [2] develops additional sufficient median and sparse testing criteria and still presents characterization directions as open; it does not contain either construction above. This is a bounded novelty audit, not an exhaustive priority claim.

References

- [1] A. K. Lerner, E. Lorist, and S. Ombrosi, *BMO with respect to Banach function spaces*, Math. Ann. **388** (2024), 4053–4082; arXiv:2204.11099. Included as `source_paper.pdf`.
- [2] S. Hashemi Sababe, *Sharp median testing and sparse criteria for generalized BMO spaces*, arXiv:2606.01688 (2026). Included as `supporting_2606.01688.pdf`.

Exact source excerpt: Questions 6.1 and 6.2

This implies (5.5) and therefore completes the proof. $\square$

6. OPEN QUESTIONS

In this final section we collect several open questions.

Question 6.1. Is there a family normalized quasi-Banach function spaces $X = \{X_Q\}$ for which the embedding $\text{BMO}_X \hookrightarrow \text{BMO}$ fails?

Observe that by Proposition 3.9 the absolute value mapping cannot be bounded on BMO_X and by Corollary 5.6 such an X cannot be rearrangement-invariant. Moreover, for every concrete, non-rearrangement-invariant family $X = \{X_Q\}$ considered in this paper, the embedding $\text{BMO}_X \hookrightarrow \text{BMO}$ holds.

If the answer to Question 6.1 is positive, i.e. the embedding $\text{BMO}_X \hookrightarrow \text{BMO}$ is nontrivial, we can ask the following question:

Question 6.2. Does the embedding $\text{BMO} \hookrightarrow \text{BMO}_X$ imply the converse embedding $\text{BMO}_X \hookrightarrow \text{BMO}$?

By Theorem 3.3, the embedding $\text{BMO} \hookrightarrow \text{BMO}_X$ provides a lot of information about $X = \{X_Q\}$, and the question is whether this information enough to establish that $\text{BMO}_X \hookrightarrow \text{BMO}$. One may also consider a stronger version of this question, asking whether the embedding $\text{BMO} \hookrightarrow \text{BMO}_X$ implies $\text{BMO}_X^* \hookrightarrow \text{BMO}$.

In Example 5.5 we established that the embedding $\text{BMO}_X^* \hookrightarrow \text{BMO}$ can fail. It is therefore natural to ask the following question: